

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2024 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

4049/01

Paper 1

20 August 2024

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total marks for this paper is 90.

Total Marks

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formula for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}bc \sin A$$

- 1 A cuboid has a volume of $(7 + 3\sqrt{5}) \text{ cm}^3$ and its base has an area of $(4 + 2\sqrt{5}) \text{ cm}^2$.
Find, **without using a calculator**, the height of the cuboid in the form $(a + b\sqrt{5}) \text{ cm}$. [3]

- 2 Express $y = 12x - 3 - 4x^2$ in the form $y = a(x + b)^2 + c$ and hence explain why the value of y is always less than or equal to 6 for all real values of x . [4]

- 3** The expression $ax^3 + bx^2 - 5x + 7$, where a and b are integers, leaves a remainder of R when divided by $x + 1$ and a remainder of $2R - 1$ when divided by $x - 2$.

(a) Show that $5a + b = 13$. [3]

(b) Given that $ab = 6$ and $b > a$, find the value of a and of b . [3]

4 Integrate $\frac{4}{x^3} + \frac{3}{3x-7}$ with respect to x .

[4]

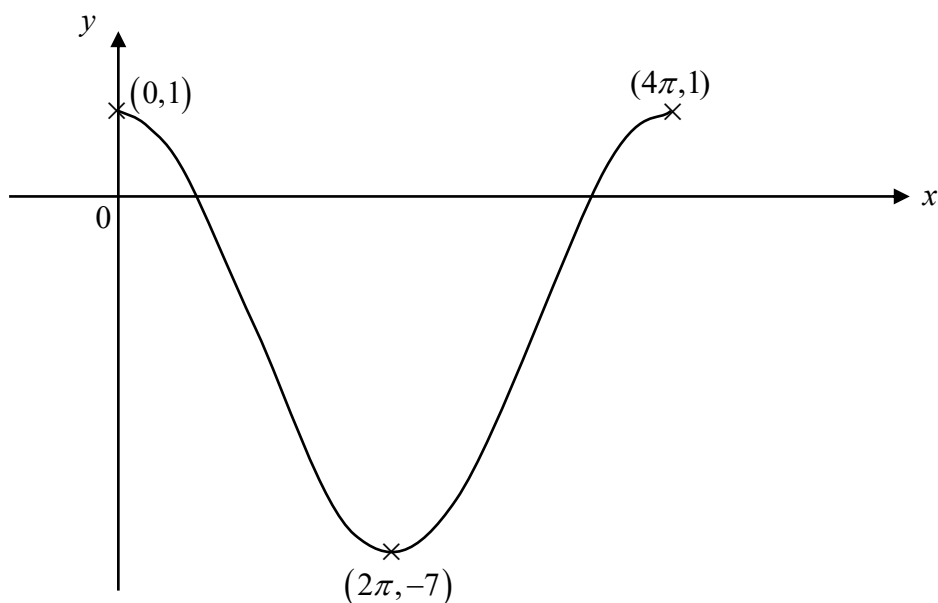
5 (a) Show that $16 \sin \frac{x}{2} \cos \frac{x}{2} - 8 \cos 2x + 9 = (4 \sin x + 1)^2$. [4]

(b) Hence solve the equation $16 \sin \frac{x}{2} \cos \frac{x}{2} - 8 \cos 2x + 9 = 9$ for $0 \leq x \leq 2\pi$, giving your answers in terms of π . [3]

- 6 Express $\frac{4x^3 + 3x^2 + 2x + 1}{4x^3 - x^2}$ in partial fractions. [6]

7 (a) Express $3\sin\theta + 4\cos\theta$ in the form $R\cos(\theta - \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. [4]

(b) Hence state the largest and smallest values of $50 - (3\sin\theta + 4\cos\theta)^2$ and find the corresponding values of θ for $0^\circ \leq \theta \leq 180^\circ$. [4]



The diagram shows part of the curve $y = a \cos bx + c$.

The curve has maximum points at $(0, 1)$ and $(4\pi, 1)$, and a minimum point at $(2\pi, -7)$.

- (a) Find the equation of the curve. [3]

- (b) The equation $a \cos bx + c + 1 = -\frac{x}{\pi}$ can be solved by drawing a suitable straight line on the diagram.

By finding the equation of the straight line, state the number of solutions of the

equation $a \cos bx + c + 1 = -\frac{x}{\pi}$ for $0 < x < 4\pi$ radians. [2]

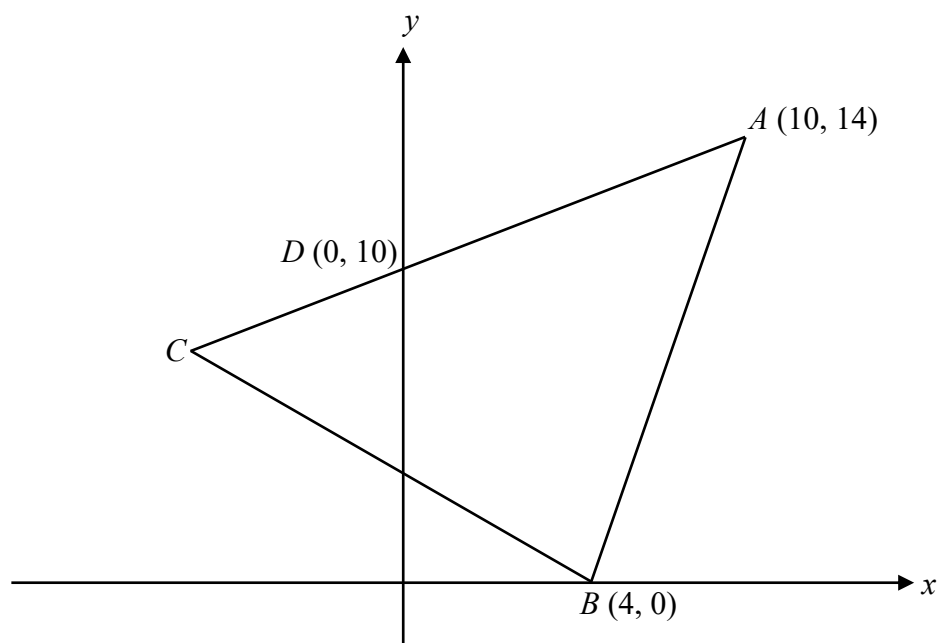
- 9 The volume, $V \text{ cm}^3$, of water in a container is given by $V = \frac{\pi h^3}{9}$, where $h \text{ cm}$ is the depth of water in the container. Due to a leakage at the bottom of the container, the depth of the water in the container decreases at a rate of $\frac{t}{4} \text{ cm/s}$, where $t \text{ s}$ is the time after the leakage started.

- (a) Given that the initial depth of water in the container is 20 cm, show that

$$h = -\frac{t^2}{8} + 20. \quad [2]$$

- (b) Hence find the rate of change of the volume of water in the container 6 seconds after the leakage started. [4]

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The diagram shows a triangle ABC with vertices $A (10, 14)$, $B (4, 0)$ and C . D lies on the line AC with coordinates $(0, 10)$.

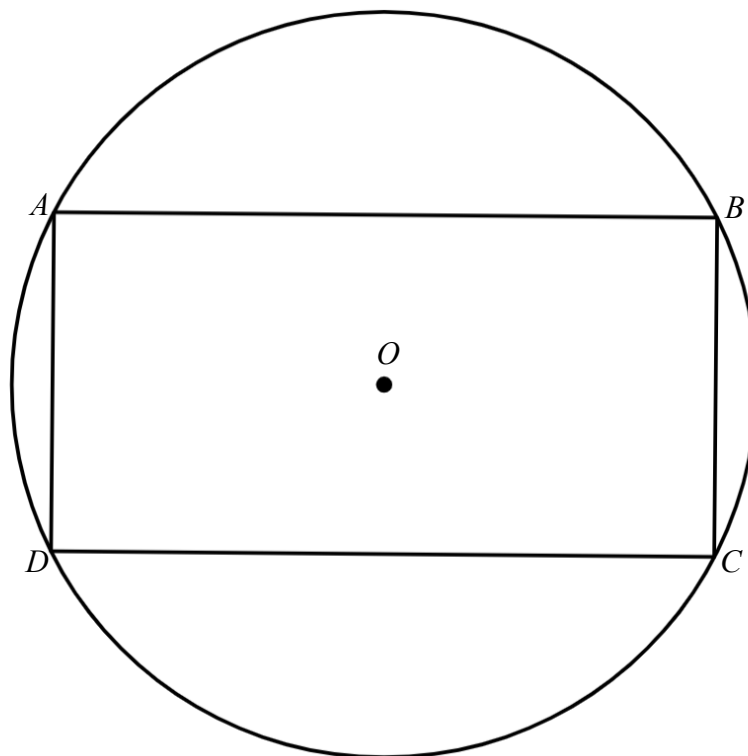
- (a) Given that $BC = \sqrt{145}$ units, find the coordinates of C .

[4]

- (b) Find the area of triangle ABC .

[2]

11



The diagram shows a circular field with centre O and radius 50 m.

A , B , C and D are points on the circumference of the field.

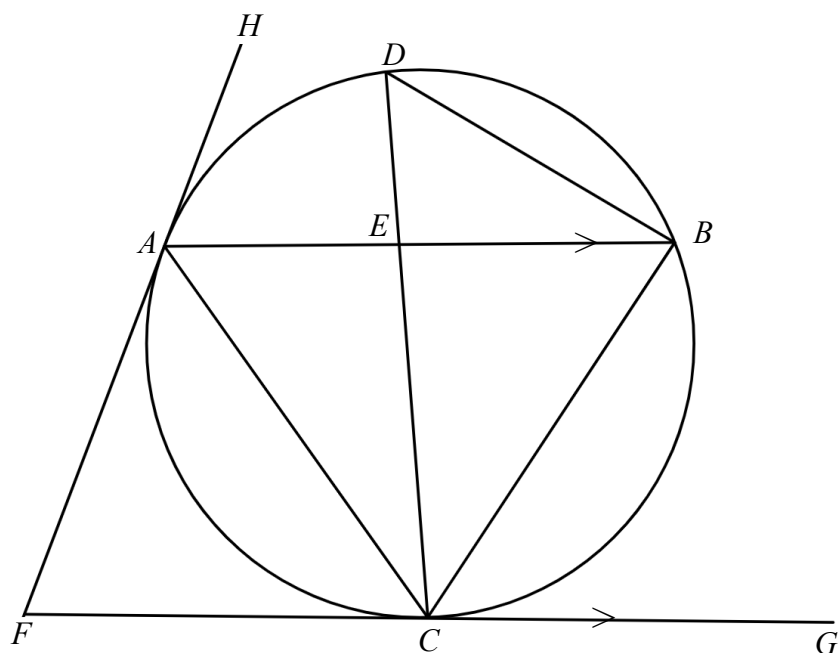
A rugby pitch is set up in the shape of a rectangle $ABCD$ such that $AB = 4x$ m and $BC = 4y$ m.

(a) Show that the area, A m², of $ABCD$ is given by

$$A = 16x\sqrt{(625 - x^2)} . \quad [3]$$

- (b) Given that x can vary, find the exact value of x for which A is stationary. [4]

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The diagram shows triangles ABC and BCD whose vertices lie on the circumference of a circle.

The chords AB and CD intersect at E and AB is parallel to FG .

The lines FAH and FCG are tangents to the circle at A and C respectively.

(a) Show that angle $CDB = \text{angle } CBE$.

[3]

(b) Show that $AC = BC$.

[3]

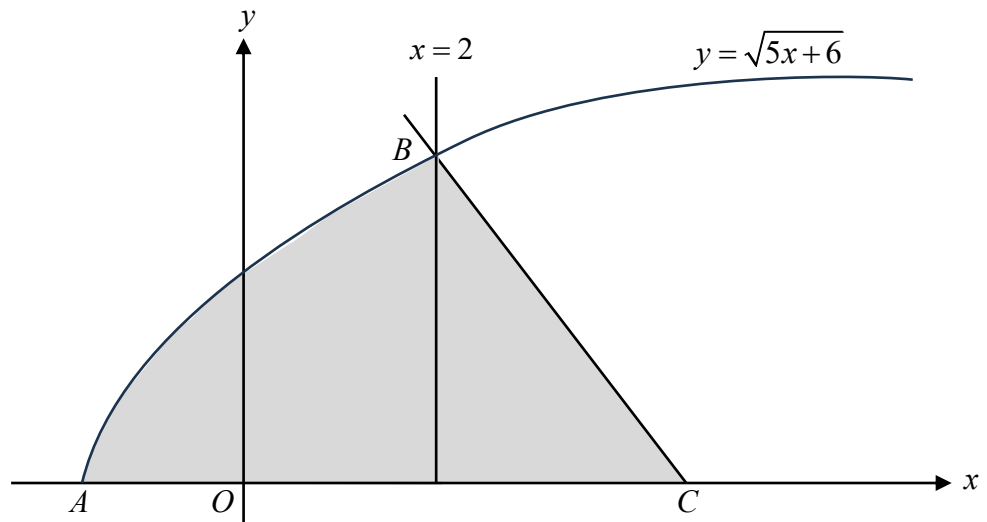
(c) Show that $\text{angle } AFC + 2 \times \text{angle } ABC = 180^\circ$.

[3]

- 13 (a) Show that the solution of the equation $\left(\frac{4}{3x}\right)^{\lg 4} = (6x)^{\lg 2}$ is $x = \frac{2}{3}$. [5]

- (b) Express the equation $\log_2 x + \log_4(x+3) = 3$ as a cubic equation in x . [4]

14



The diagram shows part of the curve $y = \sqrt{5x+6}$. The curve meets the x -axis at the point A and the line $x = 2$ at the point B . The normal to the curve at B meets the x -axis at the point C .

Find the exact area of the shaded region.

[10]

Continuation of working space for Question 14.

END OF PAPER

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